Carnegie Mellon University

Dietrich College

Information Systems

GIS ASSIGNMENT
Fall Semester, 2026

Instructions: Complete ALL THREE PARTS below.

This assignment will be graded on the correctness, completeness, attention to detail and quality of your deliverables and commentary. Remember that a bit of additional digging, narrative and exploration beyond the minimum requirements always makes a good impression! And, save your work often at convenient checkpoints; there is no undo function.

Part 1. Hurricane Evacuation Routes in Houston, Texas



Hurricanes can result in tremendous damage and loss of life. Recent major storms in 2024 across Florida and the US Southeast have been devastating and demonstrate the need for public safety planning.

Houston, Texas, among many other cities, has a long history of extreme rainfall events. It is one of America's most flood prone cities. Major flooding events are triggered by tropical storms and hurricanes, sometimes more than once in the same year. For example, Hurricane Harvey in 2017 was the most extreme flood in Houston's history. Nederland, Texas, measured 60.58 inches of rain from Harvey, which resulted in massive flooding and 68 deaths in Texas alone.

Natural disasters, such as hurricanes and flooding, often require evacuations for safety reasons. However, there is often a large portion of the population that ends up staying behind through the storm. Sometimes people choose to stay behind despite an evacuation order; other people are unable to evacuate due to many factors, such as age, disabilities, lack of transportation and illness; and sometimes officials do not issue an evacuation order at all.

For this part, you will map evacuation routes, some demographic data and flood risk zones in Houston. We will find areas where people may have difficulty evacuating in time.

Begin by Completing the Getting Started With ArcGIS Online - 'Houston Evacuation Map' exercise - ***up to, but not including '7. In the Share window, choose Everyone (public) and click save***':
https://learn.arcgis.com/en/projects/get-started-with-arcgis-online/.

(*Rather, you will share your map with the class group as indicated below.)*

Next, modify your map to more thoroughly investigate community risk (and note that this method could be used for just about any US coastal city at risk of flooding):

1. Go back to the Effects panel for the 'Pct Households with No Vehicle' layer (Drop Shadow + Transparency) and set the threshold as greater than 24.8. Thus, more census tracts appear in your map (25% or higher). Close the Effects panels.

2. To investigate the potential risk for these households in the event of a major flood, add the layer 'HarrisCountyFloodZones' by Learn ArcGIS from ArcGIS Online. (Note: No spaces in the layer name.) This layer shows the 1% flood risk - that is, the risk of a flood here is 1% every year, or, sometimes called the 'hundred-year' flood probability. However, recent natural disasters suggest the probabilities may be be substantially higher. Perhaps 100 year floods now occur every 10 years. As recently as Labor Day, 2025, for example, there was significant flooding in Houston.

3. Drag the Percent of Households with No Vehicle layer above the Flood Zones layer.

4. The color of the Flood Zones layer is similar to the color ramp for the No Vehicles layer. To make the contrast more apparent between the flood zones and the Households layer, style the Flood Zones layer (Edit layer style) with a fill color somewhere in the green, blue or aqua range. At this point, your map should be styled something like this sample section.



5. Add 'PovertyTractsHarrisCounty' from 'My organization' - author is Randy Weinberg. The 'Visibility' of this layer has been set to 'Cities' so that is it only visible when you zoom in to the city level or closer. If the layer is grayed out, zoom in.

This layer is derived from the American Community Survey (ACS) and shows the number and percentage of people living in poverty in each census tract. Each census tract is symbolized by a centroid. Thus, you can see both the percentage of

households without a vehicle and the population living in poverty on the same map. If you click on the 'Legends' icon (three below 'Layers') you can see the symbology.

6. Finally, looking at your map, can you hypothesize a relation between areas of high households without vehicles and high populations of poverty. If so, what might this suggest about risk to populations and preparedness for the next 'big one'? As examples, select a few (maybe 3) census tracts that appear to be risk prone. Areas to the southwest of your map might be the most suggestive, if at all. How might one establish this relationship or correlation?

To Turn In - Part 1

Share your map with the Organization (left hand menu column - share symbol).

Create a StoryMap to tell the story. Start by selecting **+ *Create / Story*** to start a new StoryMap. Select a style and add an interesting cover photo or image that give the viewer some context.

Your StoryMap should show an image of your final map and an interesting narrative. Describe the contents of your map and any additional commentary about what the map illustrates about potential flooding and areas at risk in the Houston area. Add any additional text, images or videos to better tell the story about the risk of flooding in Houston. Discuss your hypothesis (or lack thereof) from part 6 above. Ask, and answer your own interesting research question about the situation in Houston.

Publish your StoryMap and share with Organization and under Set group sharing, select "*CMU 67-336 F26 Part 1*". If you are prompted that an additional feature needs to be shared, go back to the original map and share the map with the Organization.

PART 2. California Wind Potential - RePower Project (USGS)

This exercise is based on the Environmental Protection Agency's (EPA's) Re-Powering America's Land: Siting Renewable Energy on Potentially Contaminated Lands, Landfills, and Mine Sites. The idea is that land that is no longer viable for habitation, agriculture, residential or commercial development may be suitable for siting solar, wind, or thermal energy generators. (See: https://www.epa.gov/re-powering)

From the site:

"EPA's RE-Powering America's Land Initiative encourages renewable energy development on current and formerly contaminated lands, landfills, and mine sites

when such development is aligned with the community's vision for the site. The Initiative identifies the renewable energy potential of these sites and provides other useful resources for communities, developers, industry, state and local governments or anyone interested in reusing these sites for renewable energy development."

In this exercise, we will look at the wind potential in a number of EPA designated sites in California. There is a file called 'EPARePoweringCA2026.xlxs' included with this assignment. This spreadsheet contains information about selected sites identified in California for wind power generation potential. Each potential site has been contaminated in some way. The spreadsheet contains information about each site, including key fields like acreage, distance to transmission lines, a link to descriptive information about the site, wind speed at 40 and 80 meters above ground, and overall wind potential.

Download and open the file. Look at the contents so you have a good idea of what it contains. Note that the 'Longitude' and 'Latitude' fields in the 'RePoweringSites' sheet locate each site precisely on any map. The Screening Criteria for potential Wind turbine site are shown in the table (also in the 'Screening Criteria' sheet):

Screening Criteria					
EPA and NREL developed the following screening criteria used to evaluate renewable energy potential. Other critical factors for siting renewable energy facilities (e.g., slope), should be considered on a site-by-site basis.					
Renewable Energy Type and Scale	**Estimated RE Project Capacity Range***	**Renewable Energy Resource Availability**	**Acreage (acres)**	**Distance to Transmission (miles)**	**Distance to Graded Roads (miles)**
Solar PV		**Global Horizontal Irradiance (kWh/m2/day)**			
Utility scale PV	≥ 5 MW	N/A	≥ 35	≤ 10	≤ 10
Distributed scale PV	< 5 MW	N/A	< 35	N/A	N/A
Off-grid PV***	N/A	N/A	N/A	N/A	N/A
Unknown acreage PV****	N/A	N/A	N/A	N/A	N/A
Wind		**Wind speed (m/s)*****			
Utility scale wind	≥ 15 MW	5.5 m/s at 80 m	≥ 1,200	≤ 5	≤ 10
Community scale wind	1.5 MW - 15 MW	5.5 m/s at 80 m	40 - 1,200	N/A	≤ 1
Facility scale wind	<1.5 MW	5.0 m/s at 40 m	0.5 - 40	N/A	N/A
Unknown acreage wind****	N/A	5.5 m/s at 80 m	N/A	N/A	N/A
Biomass		**Biomass potential within 50 miles (metric tons/year)**			
Biopower	> 10 MW	≥ 150,000 (woody)	≥ 6	≤ 10	≤ 3 road; ≤ 8 rail
Biorefinery	> 20 MMgal/year	≥ 286,000 (herbaceous)	≥ 132	N/A	≤ 3 road; ≤ 8 rail
Geothermal		**Permafrost Presence**			
Heat pump	N/A	Absence of glacier or continuous permafrost	Acreage and distance variables are not included in the prescreening.		
**MW is megawatts, MMgal/year is million gallons per year.*					
*** Transmission lines are typically defined as power lines with voltages greater than or equal to 69 kilovolts (kV).*					
****Off-grid would not be connected to the grid for the purposes of this screening criteria. Off-grid does not include no-acreage sites.*					
*****Acreage information is not readily available for these sites; this is the only category in which sites without acreage information are in*					
****** Wind speed is represented in meters per second (m/s) at heights of 40 and 80 m, as specified for each resource type.*					
N/A is not applicable.					

Some vocabulary:

- MW = Megawatt. A megawatt (MW) is equivalent to 1,000 kilowatts or 1,000,000 watts. The measurement is usually used as a means of expressing the output of power plants, or the amount of electricity that is needed to power a city. A typical coal-fired power plant produces in the region of 600 megawatts. One megawatt is equal to one million watts, so for one instant, one megawatt can power approximately 1000 homes.

- Wind speed is represented in meters per second (m/s) at heights of 60 and 80 m

- An Acre is is 43,560 square feet of land, or roughly 75% of an American football field including the end zones.

- A Kilovolt (or 1000 volts) represents the amount of electricity that can be pushed through a circuit.

Open a new map. Under the Add (+) option select "Add layer from file" and select the file called 'EPARePoweringCA2026.xlxs'. When prompted for 'Add Layer', Publish this sheet, select "RePowering Sites' and Next. Select all the fields and Next. Note that the 'Latitude' and 'Longitude' fields have been selected to locate each site. Important: Change the Title of your Layer to EPARePoweringCA2026. Then, add a summary and optional tags and add the layer to the map: 'Create and add to map'.

Style the points you have added so that they are more clearly visible on your map (enlarge, fill color, symbol, etc.) Select a base map that is helpful in visualizing this data.

Add the layer 'California County Boundaries' (May 6, 2026 or later) from ArcGIS Online. Style this layer so the Outline is Black. Create Labels for this layer as 'County Name', pick a contrasting color and turn off 'halo'. Click on the … next to the California County Boundaries layer in your layers list. Show Properties. Set visibility to Countries - Small or States/Provinces. Highlight the California County Boundaries one more time and turn off 'Enable pop-ups'. Save your work!

The most important step: Use the filter to identify three or four highly feasible sites. You might consider factors such as wind generation capacity (and note that the maximum cutoff in this dataset is 1600 MW), nearness to transmission lines, scale (utility, community, facility), acreage, location, and other factors that seem reasonable. Note your choices. If you view the EPARePoweringCA2026 table (… Show Table), you will note there are two fields in the table that may contain useful information about these sites: 'Site_Information' and 'RE_Powering_Profile'. Record useful information describing the state of the site.

To Turn In - Part 2 - A StoryMap

Select 3 (or more, as you wish) sites for analysis; zoom closely into each so you can get a more detailed view of each site, understanding the location, background of the site, the terrain, and the surrounding environment. Explore the natural and human built features of the site. For each site, note the location, features of interest (natural and human) near the site, proximity to nearest city and distance to transmission lines. Complete the following survey for each of your sites to see if any of them are actually feasible per the above criteria. Feel free to use any additional resources that might be helpful. Properly attribute your sources of information.

1. Name, location and description of each site - including previous usage or status of the site (if available). The links provided in the dataset should be helpful. Augment with any personal observations if you are familiar with the area.

2. Description of surrounding terrain and nearby communities or features of interest. Note distance to transmission lines and other relevant infrastructure. Provide informative description(s) of nearby populated places or communities or points of relevant interest.

3. Potential renewable energy type and scale, including estimated project capacity. Is there sufficient wind at 80 meters to meet minimum requirements? Is the site feasible?

4. After completing your analysis of your sites, is there one you would nominate as the most viable for location one or more wind turbines? Since site cleanup and development costs may be in the tens or hundreds of millions of dollars or more, choose only one site and explain your reasoning. You are welcome to add additional analysis (for example, through additional layers or additional research) or commentary that you think might be helpful. (You might decide none are worthy of nomination. If so, explain your reasoning.)

As above, share your map with the Organization (left hand menu column - share symbol).

Create a StoryMap with a description of the analysis, a map of California showing your sites, and three zoomed in area maps showing your sites and narrative texts. Tell a good story with your map(s) and narratives. Credit all sources.

Share your StoryMap with the Organization and with the class group *CMU 67-336 F26 Part 2.*

Part 3. Our Local Built Environment and Urban Safety Audit



The design of our streets and neighborhoods has a direct, measurable impact on human health and safety. Features like heavy traffic, busy intersections, damaged or uneven sidewalks or pavements, unprotected walkways, uncontrolled busy intersections, trip hazards and unsafe bike lanes all contribute to safety risks and health hazards. High quality built infrastructure is essential for physical activity, mobility, safety and lower stress levels.

In this assignment, you will act as an environmental health / safety inspector. You will conduct a local Urban Safety / Built Environment audit of an area within a short walk of CMU. You will map and observe traffic safety and/or mobility hazards using various dafa layers, your own senses and judgment and propose infrastructure fixes.

This part involves two activities. 1) Creating a map showing areas of risk to pedestrians, cyclists and vehicular traffic, and 2) Taking first-hand observations of 3 notable intersections, crossings or particular points of interest relevant to the topic.

Instructions

Activity 1

Log into ArcGIS Online and open the Map Viewer. Add the following layers from ArcGIS Online:

1. *High Injury Network owner:domi_leonardg*

The City's High Injury Network (HIN) identifies areas with high frequencies of traffic-related injuries and fatalities.

2. *CrashData HIN owner:domi_leonardg*

Hot spots of all traffic collisions 2017-2022 are shown ('HIN' = High Injury Network)

3. *Ped _ Bike Crashes _18 _ _22 owner:domi_leonardg*

Collisions with pedestrians, 2018-2022, and bicycles are shown

In Map Viewer, zoom in to the area around CMU, showing key 'hot spots' and numerous ped/bike crashes. Note that the hot spots indicate more than one incident involving vehicle collisions and/or collisions with pedestrians or bikes. For examples, the hot spot at Forbes Avenue and Morewood represents approximately 14 incidents; the hot spot at Neville and 5th Avenue represents approximately 22 incidents. (Also the CrashData HIN data includes pedestrian and bike incidents that may also be shown in the Ped_Bike_Crashes data layer.) If you look at the tables, you can see full details.

Consider (but not required) selecting a different basemap than the default to provide better contrast and detail (for example, one of the 'street' basemaps.) Set a bookmark so you can easily return to your map.

Save your work.

Activity 2

Select three (or more) risky sites within a reasonable distance from campus or your residence (say, 10 to 15 minutes walking radius).

At each of these locations, record:

1. The exact location (note the cross streets, intersection, or address).
2. A qualitative observational note detailing what you experienced in relation to traffic, pedestrian or bicycle conditions or hazards. (*Example: "There have been several collisions at this intersection. This intersection could be dangerous to pedestrians because … In particular, for people with mobility issues, there is insufficient time to cross safely. Further, the the sidewalk corners are all raised with no curb ramps."*)
3. One original photograph looking directly at each location in question. (Important: Save or export your photographs so they are less than 1MB in size; they will load much faster in the next part).

Mapping Your Observations (ArcGIS Online)

Log into ArcGIS Online and open the **Map Viewer in one tab and your Content in a second tab. In the Content tab, upload your reduced sized photographs, one at a time. For each photo, select Sharing and share with Everyone** (third option from right hand … column). If you click on the … symbol, you will see properties, including the URL to your photo. **Keep this tab open for now.**

Next, in Map Viewer, Zoom in to your route and add your locations. On the right hand toolbar, click **Sketch.** The Sketch pane appears, and the sketch tools appear on the map. From the sketch tools, choose **Draw a point.** Click a point on the map where you want to place it. A default point is placed on the map. Go back to the vertical pane and click on the top option (select feature). This turns off the automatic (and annoying) accidental drawing of additional points. Then go back and click on your point. The Sketch pane appears. Give the point a clear name like *5th Avenue and Craig Street* (select Feature Name). If you wish, you can change the symbol and color of the point.

Configure a pop-up for this point. From the Sketch pane, select Pop-up. Then, Enable Pop-ups. Add a Title and add Text with your narrative about the point. To add your photo, select Add Content / Image. In the Content tab, select your photo (already shared with Everyone) and copy the URL. Paste the URL into the URL field. Your photo should appear in your text box. Add a title and/or caption if you wish. Then Done. Complete for all three of your sites.

Finally, Create and publish an ArcGIS StoryMap
Open **ArcGIS StoryMaps** in a new tab and create an interactive report.

Your story must include:

- **Title & Intro:** Add a descriptive title and, perhaps a subtitle. Briefly explain the purpose of your study. Select a style and add an interesting cover image. Add any further images and/or text that give the viewer some additional context.
- **The Interactive Map:** Embed your saved Web Map (zoom appropriately to your journey) into the story so the reader can click on your points and view your field text and photos.
- **Critical Reflection:** Write a summary of your findings and imagine you are an urban designer with a limited budget. What is **one immediate intervention** you would make based on one or more of the points you mapped (e.g. fixing a trip hazard, adding a speed hump, increasing pedestrian crossing time?) Which point you would fix first?

As above, share your map with the Organization (left hand menu column - share symbol).

Share your StoryMap with the Organization and with the class group *CMU 67-336 F26 Part 3*.